MILITARY SPECIFICATION

DIVING APPARATUS, SELF CONTAINED DEMAND TYPE

1. SCOPE

1.1 Scope. - This specification covers diving apparatus, demand type for use to a depth of 200 feet without lines to surface.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-M-15071 - Manual, Service (Instruction Books) For Shipboard Electrical and Mechanical Equipment.
MIL-P-15137 - Provisioning and Technical Documentation For Repair Pairs for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-C-16310 - Cylinders, Compressed Gas (Compressed Air and Carbon Dioxide, Nonshatterable and Nonmagnetic).
MIL-R-19558 - Regulator, Air Demand, Diver's.
MIL-M-19595 - Magnetic Effect Limits for Nonmagnetic Equipment and Metals (Special Purpose).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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DRAWINGS

AIR FORCE - NAVY AERONAUTICAL
AND 10050-12 - Bosses, Standard Dimensions for Gasket Seal Straight Thread.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Applications for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standard for Federal Service.

(Application for copies should be addressed to the Superintendant of Documents, Government Printing Office, Washington 25, D. C.)

ALUMINUM ASSOCIATION STANDARDS

(Application for copies should be addressed to the Aluminum Association, 420 Lexington Avenue, New York 17, N. Y.)

3. REQUIREMENTS

3.1 Preproduction sample. - Prior to the start of production, a preproduction sample shall be submitted for testing in accordance with 4.2.

3.2 Material. - Material shall be of the best quality and entirely suitable for the intended purpose. Material shall meet the magnetic effect limits as specified in Specification MIL-M-19595.

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3.3 General requirements. - The apparatus shall be suitable for use by personnel operating without connections to the surface and shall provide sufficient air for depths to 200 feet. The apparatus shall consist of mouthpiece and hose assembly, regulator, manifold with low pressure warning device, cylinders, cylinder strap, harness and components necessary for complete assembly.

3.4 Detail requirements. -

3.4.1 Mouthpiece and hose assembly. - The mouthpiece and hose assembly shall be a commercial type suitable for the purpose intended. The assembly shall include inhalation and exhalation check valves. When in use, the exhalation valve shall be located on the left side and the inhalation valve on the right side.

3.4.2 Regulator. - The regulator shall be in accordance with Specification MIL-R-19558.

3.4.2.1 Cylinders. - Cylinders made from tubing with spun ends shall be made from 6061-T6 aluminum in accordance with Aluminum Association Standards. Cylinders made by the deep drawing process shall be made from 14S-T6 aluminum in accordance with Aluminum Association Standards.

3.4.2.2 Size. - The cylinders shall be 7-1/2 inches in outside diameter (OD) and 25-1/4 inches long.

3.4.2.3 Ends. - In forming the ends sufficient material shall be provided to face off the end to form a 1-5/8 inch minimum flat.

3.4.2.4 Threading. - Cylinders shall be tapered at one end with a 3/4 inch 14 NPSM thread in accordance with Handbook H28. The thread opening shall be machined for an "O" ring seal in accordance with Drawing AND 10050-12. There shall be no folds, cracks or other imperfection after threading.

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3.4.2.5 Pressure rating. - The cylinder shall be designed for a working pressure of 3000 pounds per square inch (psi) and a test pressure of 5000 psi.

3.4.3 Manifold. - The manifold shall be a three section type. The two end plugs shall have matching threads and "O" ring fittings for the cylinders. One plug shall have the low pressure warning indicator. The indicator shall warn the diver when the air pressure in the cylinder reaches 300 psi. Safety plugs shall be provided in end sections of the manifold. The center section shall have the valve and fittings for attaching the regulator.

3.4.4 The entire apparatus when the cylinders are charged to 1500 psi shall be neutral in water and shall have proper trim characteristics for swimming.

3.5 Repair parts. - The bidder shall submit a list of recommended repair parts for the approval of the bureau or agency concerned. The repair parts shall be in accordance with Specification MIL-R-15137.

3.6 Service manuals and repair parts book. - Two complete service manuals and repair parts books shall be furnished with each diving apparatus. The service manual shall be in accordance with class B of Specification MIL-M-15071. A preliminary rough draft copy of the service manual shall be submitted to the bureau or agency concerned for approval.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. - Prior to the start of production one equipment shall be submitted to the test of 4.2.1.

4.2.1 Swimming test. - Using the arrangement shown on figure 1 and swimming against a 12 pound pull it shall be possible for the diver to obtain sufficient air for breathing at any depth to 200 feet without excessive breathing resistance. The diver shall swim for a period of 30 minutes. During the swimming test the pressure on the high pressure side will be varied from 3,000 psi down to 300 psi.

4.3 Sampling for lot acceptance. -

4.3.1 Lot. - All parts of each kind offered for delivery at one time shall be considered a lot.

4.3.2 Sampling for examination. - Sample units of each kind shall be selected in accordance with inspection level III of Standard MIL-STD-105 for the examination of 4.4. Any unit having one or more defects shall be rejected. The AQL shall be 1.5 percent.

4.3.3 Sampling for hydrostatic pressure test. - Each cylinder and manifold shall be tested as specified in 4.5.1.

4.3.4 Sampling for magnetic effect test. - One test specimen shall be cut from each sheet of metal and each coil of wire for the test specified in 4.5.2. Fabricated parts shall be sampled and tested in accordance with Specification MIL-M-19595.

4.3.5 Sampling for low-pressure warning device test. - Sample units shall be selected in accordance with inspection level III of Standard MIL-STD-105 and tested as specified in 4.5.3. If any unit fails, the lot shall be rejected.

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4.4 Visual and dimensional examination. - Each sample mouthpiece, valve, hose, pressure indicator, cylinder, manifold, and harness shall be examined and measured to verify conformance with all requirements of this specification which do not involve tests.

4.4.1 Cylinder and manifold threads shall be checked by means of "GO" "NO-GO" gages specified in the sub-section pertaining to gages in Handbook H28 and supplement thereto.

4.5 Tests. -

4.5.1 Hydrostatic pressure. - All cylinders and manifolds shall be tested, either individually or as assemblies to a 5000 psi hydrastatic pressure. Any leakage shall be cause for rejection. The permanent expansion shall not be more than ten percent of the total expansion.

4.5.2 Magentic effect. - The magnetic effect limits shall be determined in accordance with Specification MIL-M-19595.

4.5.3 Low pressure warning device. - The low pressure warning device shall be tested by subjecting the manifold to 300 to 350 psi air pressure. The flow shall be satisfactory for swimming at pressure when cylinder pressure is above 350 psi. When the pressure is between 300 to 350 psi the air flow shall be stopped. Minor leakage at 300 psi is not cause for rejection.

5. PREPARATION FOE DELIVERY

5.1 Domestic shipment and early equipment use and storage of onboard repair parts. -

5.1.1 Basic equipment. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early use. Manuals (see 3.6) shall be enclosed and sealed in waterproof bags or envelopes.

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5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during shipment from the supply source to the using activity for early use. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. -

5.1.2.1 Preservation and packing. -

5.1.2.1.1 Parts subject to corrosion or deterioration shall be unit protected and packaged in accordance with method IC or IA of Specification MIL-P-116. Selection of the method and sub-method for the part concerned shall be at the option of the contractor and in accordance with the guide lines of Specification MIL-P-116.

5.1.2.1.2 Parts inherently resistant to corrosion or deterioration shall be unit protected and packaged in bags, cartons or other containers in accordance with method III of Specification MIL-P-116. Bags shall be closed with adhesives, tape or by heat sealing. Closure of cartons or other containers shall be strong enough to permit handling of the container and contents without failure.

5.1.2.2 Packing. - Onboard repair parts shall be packed for shipment in accordance with 5.1.1.2.

5.1.2.3 Marking. - Shipment markings shall be in accordance with Standard MIL-STD-129, level A markings for preservation and packaging, level C markings for packing.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

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(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment which may be required when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Level A. -

5.2.1.1.1.1 Components. - The components necessary for complete assembly (see 3.3) shall be packaged in accordance with method III of Specification MIL-P-116. All openings to the component parts shall be sealed with plugs or pressure sensitive tape. Component parts such as cylinder block assembly and demand regulator assembly shall be unit packaged in fiberboard boxes conforming to class 2 of Specification PPP-B-636. Type and style of box shall be at the option of the contractor. Blocking, bracing and cushioning shall be provided to attached parts (manifold, valves and so forth) to prevent damage during handling, shipment and storage. Box closure, and sealing of all center and edge seams and manufacturer's joint with pressure sensitive tape shall be in accordance with the appendix to the box specification.

5.2.1.1.1.2 Repair parts. -

5.2.1.1.1.2.1 Parts subject to corrosion or deterioration shall be unit protected and packaged in accordance with method IC or IA of Specification MIL-P-116. Unless otherwise specified in the contract or order the selection of the method and sub-method for the part concerned shall be at the option of the contractor.

5.2.1.1.1.2.2 Parts inherently resistant to corrosion or deterioration shall be unit protected and packaged in bags, cartons or other containers in accordance with method III of Specification MIL-P-116. Bags shall be closed with adhesive, tape or by heat sealing. Closure of cartons or other containers shall be strong enough to permit handling of the container and contents without failure.

5.2.1.1.2 Level C. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early use.

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5.2.1.2 Packing. -

5.2.1.2.1 Level A. -

5.2.1.2.1.1 Each complete assembly (see 3.3) shall be individually packed, or repair parts (see 3.5) shall be packed in containers conforming to any of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-576	Class 2
PPP-B-585	Class 3 use
PPP-B-591	Overseas type
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Class 2
PPP-B-640	Class 2

Boxes shall be closed, strapped or banded in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight of the applicable box specification.

5.2.1.2.1.2 Caseliners. - Shipping containers shall be provided with closed and sealed caseliners in accordance with Specification MIL-L-10547 and appendix thereto. Caseliners will not be required when shipping containers are packed with packages meeting the following:

(a) Method IA or IC packages.
(b) Unit packages for which the container is a fiberboard box conforming to class 2 of Specification PPP-B-636 and all center and edge seams and manufacturer's joint are waterproofed with pressure sensitive tape in accordance with the appendix to Specification PPP-B-636.
(c) When the shipping container conforms to class 2 of Specification PPP-B-636 or PPP-B-640 and waterproofed as specified in (b) above.

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5.2.1.2.2 Level B. - Each complete assembly (see 3.3) shall be individually packed, or repair parts (see 3.5) shall be packed in containers conforming to any of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-576	Class 1
PPP-B-585	Class 1 or 2 use
PPP-B-591	Domestic type
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-636	Class 1
PPP-B-640	Class 1

Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.1.3 Marking. - In addition to any special marking required, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

5.2.1.4 Manuals. - Manuals accompanying equipment (see 3.6) shall be packaged in accordance with method IC-3 of Specification MIL-P-116. Bulk manuals shall be packed for the level specified (see 6.1) in accordance with Specification MIL-M-15071.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable preservation, packaging, packing and marking if other than as specified in 5.1 (see 5.2).

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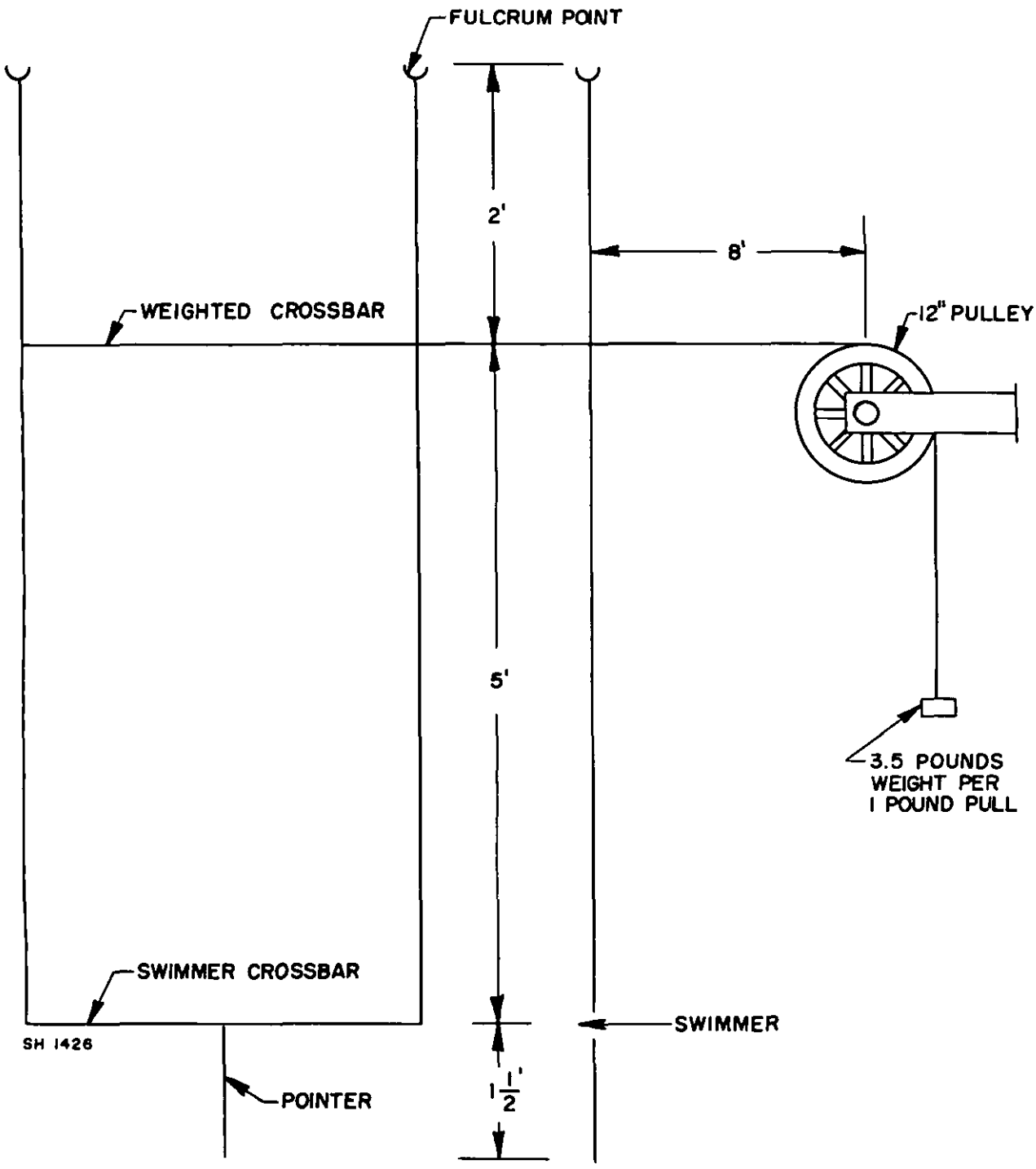


Figure 1 - Simulated swimming test arrangement.

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SPECIFICATION ANALYSIS SHEET

Instructions

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D. C.

Specification

Organization	City	State

Contract No.

Quantity of Items Procured	Dollar Amount $

Material procured under a direct Government contract ☐ or a subcontract ☐

1. Has any part of the specification created problems or required interpretation in procurement?
   a. Give paragraph number and wording.

   b. Recommendations for correcting the deficiencies.

2. Comment on any specification requirement considered too rigid.

3. Is the specification restrictive? ☐ Yes ☐ No If the answer is "Yes", in what way?

4. Remarks (Attach any pertinent data which may be of use in improving this specification.)
Place this form and papers in an envelope and send to the Bureau.

Submitted by (Print name and activity)	Date

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